

DATA DICTIONARY

Airbnb NYC Listings Analytics

Power BI End-to-End YouTube Demo Project

All Tables · All Columns · All Measures | v1.0 | May 2026

Overview

This Data Dictionary defines every column and measure in the Airbnb NYC Analytics Power BI project. Use this document as the authoritative reference when building the report, writing DAX, or explaining the dataset on the YouTube demo.

Symbol	Meaning	Symbol	Meaning
PK	Primary Key — unique identifier for the table	FK	Foreign Key — relates to another table
⚠️ DIRTY	Column has intentional data quality issue — fix required in Power Query	PQ Type	Target data type to set in Power Query after cleaning

1. fact_bookings — Fact Table

100,200 rows (incl. 200 duplicates to remove). Central transaction table. Grain = 1 row per booking.

Column Name	Raw Type	PQ Type	Key	Nullable	Example Value	Description & Business Rules
booking_id	Text	Text	No	PK	BK000001	Unique booking identifier. Format: BK followed by 6 digits. ⚠️ 200 duplicate rows exist — fix: Table.Distinct(Source, {"booking_id"})
listing_id	Text	Text	No	FK	L001	Foreign key to dim_property[listing_id]. 80 distinct values expected.
host_id	Text	Text	No	FK	H012	Foreign key to dim_host[host_id]. 50 distinct values expected.
date_key	Text	Text	No	FK	20230415	Foreign key to dim_date[date_key]. Format: YYYYMMDD as text. Do NOT convert to date — relationship uses text key.

Column Name	Raw Type	PQ Type	Key	Nullable	Example Value	Description & Business Rules
location_id	Text	Text	No	FK	LOC07	Foreign key to dim_location[location_id]. 20 distinct values expected.
price  DIRTY	Text	Decimal Number	No		150 or "\$1,500.00"	 DIRTY: ~3,000 rows stored as text e.g. '\$1,234.00'. Fix: Text.Remove(\$,comma) then Number.From. After fix, set to Decimal Number.
minimum_nights  DIRTY	Integer	Whole Number	No		2	 DIRTY: ~800 rows have value 0 or negative. Valid range: 1–365. Fix: Table.SelectRows(Source, each [minimum_nights] >= 1)
number_of_reviews	Integer	Whole Number	No		48	Total cumulative reviews for this listing at time of booking. Range: 0–500.
review_score  DIRTY	Decimal	Decimal Number	No		4.72	 DIRTY: ~1,500 rows have score > 5.0 (e.g. 5.3). Valid range: 0.0–5.0. Fix: Table.SelectRows(Source, each [review_score] <= 5.0)
availability_365  DIRTY	Integer	Whole Number	No		120	 DIRTY: ~600 rows have value 999 (sentinel error). Valid range: 0–365. Fix: Table.ReplaceValue(Source, 999, null, ...)
calculated_revenue	Integer	Decimal Number	No		1800	Pre-calculated: price × minimum_nights. Used as primary revenue measure. May need recalculation if price was dirty.
cancellation_policy	Text	Text	No		flexible	One of: flexible, moderate, strict, super_strict_30, long_term. Used in Cancellation Rate measure.
booking_source	Text	Text	No		Airbnb App	Channel source: Direct, Airbnb App, Web, Partner, Corporate. Used for booking channel analysis.

2. dim_host — Host Dimension

50 rows. Describes each Airbnb host. 3 null names and 3 malformed response rates.

Column Name	Raw Type	PQ Type	Key	Nullable	Example Value	Description & Business Rules
host_id	Text	Text	No	PK	H001	Unique host identifier. Format: H followed by 3 digits. 50 rows total.
host_name  DIRTY	Text	Text	Yes		Sarah Mitchell	 DIRTY: 3 rows are null/empty. Fix: Table.ReplaceValue(Source, null, "Unknown Host", Replacer.ReplaceValue, {"host_name"})

Column Name	Raw Type	PQ Type	Key	Nullable	Example Value	Description & Business Rules
host_since	Text	Date	No		2015-03-12	Date host registered on platform. Format: YYYY-MM-DD. Change type to Date in Power Query.
host_response_rate ⚠️ DIRTY	Text	Percentage	No		85% or 0.85	⚠️ DIRTY: 3 rows stored as decimal (0.85) instead of '85%'. Fix: detect if value < 1, then multiply by 100 and format as percentage. Final type: Percentage.
superhost_flag	Text	Text	No		t	Superhost status: 't' = Superhost, 'f' = Regular. Used in Superhost Revenue % measure.
host_listings_count	Integer	Whole Number	No		12	Total number of active listings this host manages. Range: 1–25.

3. dim_property — Property Dimension

80 rows. Describes each listing. 5 null bedroom values for studio-type properties.

Column Name	Raw Type	PQ Type	Key	Nullable	Example Value	Description & Business Rules
listing_id	Text	Text	No	PK	L001	Unique listing identifier. Format: L followed by 3 digits. 80 rows total.
listing_name	Text	Text	No		Cozy Manhattan Studio	Display name of the listing. Generated from adjective + noun combinations.
room_type	Text	Text	No		Entire home/apt	One of: Entire home/apt, Private room, Shared room, Hotel room. Used in donut chart slicer.
property_type	Text	Text	No		Apartment	One of: Apartment, House, Loft, Condo, Townhouse, Guesthouse, Villa, Studio, Hostel, Boutique Hotel.
accommodates	Integer	Whole Number	No		4	Maximum guest capacity. Range: 1–10. Used in RevPAR denominator calculation.
bedrooms ⚠️ DIRTY	Integer	Whole Number	Yes		2	⚠️ DIRTY: 5 null values (studio-type units). Fix: Table.ReplaceValue(Source, null, 0, ...) — 0 represents studio.
bathrooms	Decimal	Decimal Number	No		1.5	Number of bathrooms. Half-bath increments used (1.0, 1.5, 2.0, 2.5, 3.0).
amenities_count	Integer	Whole Number	No		32	Count of amenities offered. Range: 8–60. Higher values typically correlate with higher pricing.
host_id	Text	Text	No	FK	H001	Foreign key to dim_host. Used to link property to its host profile.
location_id	Text	Text	No	FK	LOC01	Foreign key to dim_location. Determines neighbourhood for geographic analysis.

Column Name	Raw Type	PQ Type	Key	Nullable	Example Value	Description & Business Rules
base_price	Integer	Whole Number	No		150	Reference nightly price for this listing. Used in dim_property only — actual booking price is in fact_bookings[price].

4. dim_location — Location Dimension

20 rows. NYC neighbourhood lookup table. 2 latitude values with whitespace.

Column Name	Raw Type	PQ Type	Key	Nullable	Example Value	Description & Business Rules
location_id	Text	Text	No	PK	LOC01	Unique location identifier. Format: LOC followed by 2 digits. 20 rows total.
neighbourhood	Text	Text	No		Manhattan	NYC neighbourhood name. 20 distinct values covering all 5 boroughs plus key sub-districts.
city	Text	Text	No		New York	Always 'New York' in this dataset (single-city scope).
state	Text	Text	No		NY	US state abbreviation. Always 'NY'.
country	Text	Text	No		United States	Always 'United States' in this dataset.
latitude ⚠️ DIRTY	Text	Decimal Number	No		40.7831	⚠️ DIRTY: 2 rows have leading/trailing spaces e.g. ' 40.8116 '. Fix: Text.Trim then change type to Decimal Number.
longitude	Text	Decimal Number	No		-73.9712	Decimal longitude for map visual. Negative = West of meridian (standard for NYC).
zipcode	Text	Text	No		10024	US ZIP code. Keep as Text type to preserve leading zeros.

5. dim_date — Date Dimension

1,127 rows (Jan 2023–Jan 2026). Month name casing is inconsistent — fix with Text.Proper.

Column Name	Raw Type	PQ Type	Key	Nullable	Example Value	Description & Business Rules
date_key	Text	Text	No	PK	20230415	Surrogate key in YYYYMMDD format as text. Must match fact_bookings[date_key] exactly.
full_date	Text	Date	No		2023-04-15	Actual date value. Change type to Date. Mark this column as the Date Table key.
year	Integer	Whole Number	No		2023	Calendar year. Range: 2023–2026.
quarter	Integer	Whole Number	No		2	Calendar quarter (1–4). Derived from month.

Column Name	Raw Type	PQ Type	Key	Nullable	Example Value	Description & Business Rules
month_num	Integer	Whole Number	No		4	Calendar month number (1–12). Use this to sort month_name in visuals.
month_name ⚠️ DIRTY	Text	Text	No		April	⚠️ DIRTY: Inconsistent casing e.g. 'january', 'February', 'march'. Fix: Text.Proper(each [month_name])
week_num	Integer	Whole Number	No		15	ISO week number (1–53). Useful for weekly trend slicers.
day_of_week	Integer	Whole Number	No		6	Day of week number (1=Mon, 7=Sun). Use for weekend/weekday analysis.
day_name	Text	Text	No		Saturday	Full day name. Useful for day-of-week patterns in bookings.
is_weekend	Text	Text	No		Yes	Yes/No flag. Weekend = Saturday or Sunday. Use in conditional visuals.
is_holiday	Text	Text	No		No	Yes/No flag for US public holidays. Holidays show higher booking rates in analysis.

6. DAX Measures — Measures Table

All 23 measures live in the ' Measures' calculated table (DAX expression = BLANK()). Never place measures inside fact or dimension tables.

Measure Name	Display Folder	DAX Pattern	Format	Description & Notes
Total Revenue	 Core Revenue KPIs	SUM(fact_bookings[calculated_revenue])	\$#,0.00	Sum of pre-calculated revenue across all bookings. Primary KPI for the report.
Total Bookings	 Core Revenue KPIs	COUNTROWS(fact_bookings)	#,0	Count of all booking rows after Power Query deduplication. Denominator for many rate measures.
Avg Nightly Rate	 Core Revenue KPIs	AVERAGEX(fact_bookings, fact_bookings[price])	\$#,0.00	Average price per night using AVERAGEX to correctly handle row context. Context-aware with slicers.
RevPAR	 Core Revenue KPIs	DIVIDE([Total Revenue], SUMX(dim_property, accommodates * 365))	\$#,0.00	Revenue Per Available Room. Industry KPI = Total Rev ÷ (capacity × days). Measures yield efficiency.
Avg Revenue Per Booking	 Core Revenue KPIs	DIVIDE([Total Revenue], [Total Bookings])	\$#,0.00	Average booking value. Lower than Avg Nightly Rate × Avg Nights due to mix effect.
Occupancy Rate	 Occupancy	DIVIDE(365 - AVERAGE(fact_bookings[availability_365]), 365)	0.0%	Proxy occupancy = (365 - avg days available) / 365. Higher = more booked. Null avail rows excluded.
Cancellation Rate	 Occupancy	DIVIDE(CALCULATE([Total Bookings], cancellation_policy='strict'), [Total Bookings])	0.0%	% of bookings under strict cancellation policy. Higher = less flexible inventory.

Measure Name	Display Folder	DAX Pattern	Format	Description & Notes
Avg Nights Per Booking	 Occupancy	AVERAGE (fact_bookings[minimum_nights])	0.0	Average length of stay in nights. Filtered to ≥ 1 in Power Query, so no invalid rows affect this.
Revenue PM	 Time Intelligence	CALCULATE([Total Revenue], DATEADD(dim_date[full_date], -1, MONTH))	\$#,0.00	Prior month total revenue using DATEADD. Requires dim_date marked as Date Table.
MoM Revenue Growth %	 Time Intelligence	DIVIDE([Total Revenue] - [Revenue PM], [Revenue PM])	+ 0.0%	Month-over-month growth %. Returns blank when no prior month exists (e.g. Jan 2023). Uses VAR pattern.
Revenue PY	 Time Intelligence	CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(dim_date[full_date]))	\$#,0.00	Prior year revenue for same period. Requires a contiguous date selection for SAMEPERIODLASTYEAR.
YoY Revenue Growth %	 Time Intelligence	DIVIDE([Total Revenue] - [Revenue PY], [Revenue PY])	+ 0.0%	Year-over-year growth %. Returns blank for 2021 (no prior year data in dataset).
Revenue YTD	 Time Intelligence	CALCULATE([Total Revenue], DATESYTD(dim_date[full_date]))	\$#,0.00	Cumulative revenue from Jan 1 to last selected date. Resets each calendar year.
Avg Review Score	 Review & Quality	AVERAGE (fact_bookings[review_score])	0.00	Mean guest review score. Filtered to ≤ 5.0 in Power Query. Context-sensitive with property/host slicers.
Total Reviews	 Review & Quality	SUM(fact_bookings[number_of_reviews])	#,0	Total review count across all bookings in filter context. Not distinct — sums cumulative review counts.
High Rating %	 Review & Quality	DIVIDE(CALCULATE([Total Bookings], review_score $\geq$ 4.5), [Total Bookings])	0.0%	% of bookings with review score 4.5 or above. Indicates high-quality inventory share.
Superhost Revenue	 Host Performance	CALCULATE([Total Revenue], dim_host[superhost_flag]='t')	\$#,0.00	Revenue from superhosts only. Uses relationship filter on dim_host[superhost_flag].
Superhost Revenue %	 Host Performance	DIVIDE([Superhost Revenue], [Total Revenue])	0.0%	Superhost share of total revenue. Useful KPI for marketplace quality analysis.
Superhost Avg Review	 Host Performance	CALCULATE([Avg Review Score], dim_host[superhost_flag]='t')	0.00	Average review score specifically for superhost-managed listings.
Total Active Hosts	 Host Performance	DISTINCTCOUNT(fact_bookings[host_id])	#,0	Count of unique hosts with at least one booking in the current filter context.
MoM Arrow Label	 KPI Formatting	VAR + FORMAT with ▲/▼ arrows	Text	Dynamic text label combining direction arrow with % value. e.g. '▲ 12.4% MoM'. Use as secondary KPI card value.
YoY Arrow Label	 KPI Formatting	VAR + FORMAT with ▲/▼ arrows	Text	Same pattern as MoM Arrow Label but for YoY comparison. e.g. '▼ 3.2% YoY'.
Revenue Label	 KPI Formatting	IF Revenue $\geq$ 1M $\rightarrow$ \$X.XM, else \$X.XK	Text	Compact formatted label e.g. '\$4.2M'. Used in card visuals for space-efficient display.

7. Relationship Map

All relationships are single-direction, one-to-many. Cross-filter direction = Single (Dim → Fact).

Dimension Table	Dim Key	Fact Table	Fact FK	Filter Direction
dim_date	date_key	fact_bookings	date_key	Single (to Fact)
dim_host	host_id	fact_bookings	host_id	Single (to Fact)
dim_property	listing_id	fact_bookings	listing_id	Single (to Fact)
dim_location	location_id	fact_bookings	location_id	Single (to Fact)

8. Data Quality Summary

10 total data quality issues across 5 files. All are intentional and fixable in < 5 minutes of Power Query work.

File	Column	Affected Rows	Fix Summary
fact_bookings	booking_id	200 dupes	Table.Distinct on booking_id column
fact_bookings	price	~3000 rows	Text.Remove dollar and comma, then to Decimal
fact_bookings	review_score	~1500 rows	Filter rows where review_score <= 5.0
fact_bookings	minimum_nights	~800 rows	Filter rows where minimum_nights >= 1
fact_bookings	availability_365	~600 rows	Replace value 999 with null
dim_host	host_name	3 rows	Replace null with Unknown Host string
dim_host	host_response_rate	3 rows	Detect decimal format, multiply by 100
dim_property	bedrooms	5 rows	Replace null with 0 for studio units
dim_location	latitude	2 rows	Text.Trim then convert to Decimal Number
dim_date	month_name	730 rows	Text.Proper to standardise all month casing